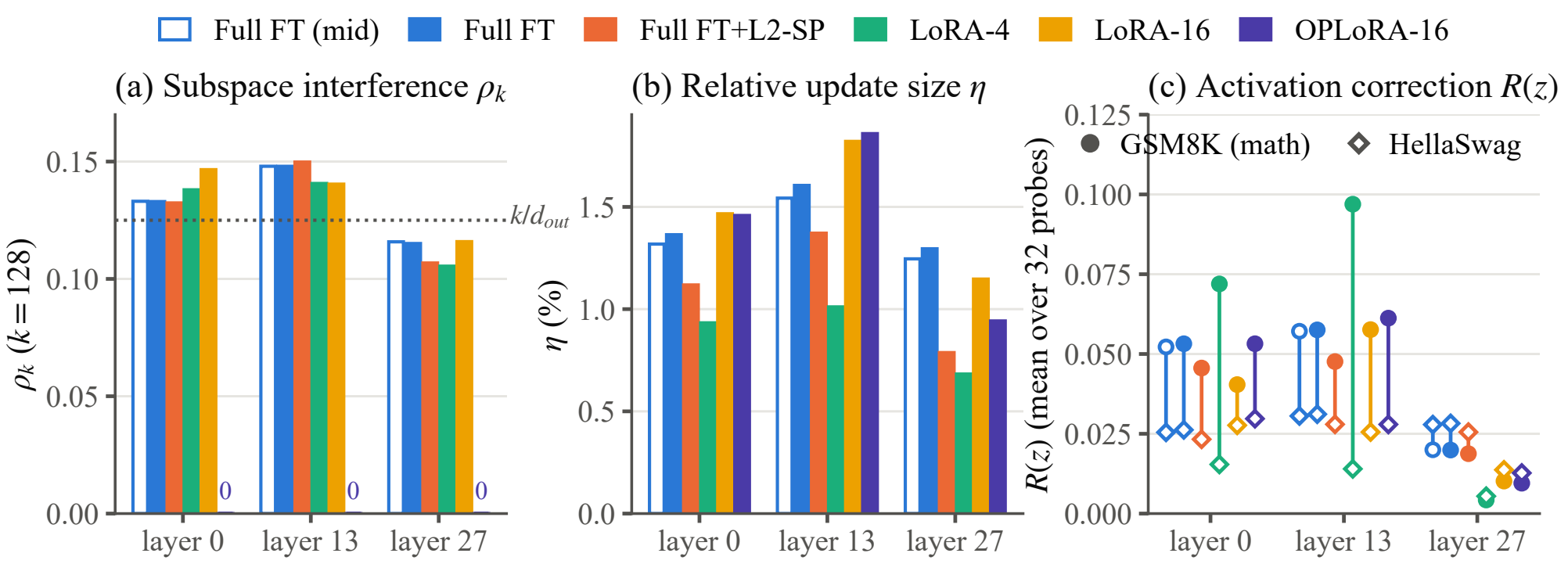
Full FT (mid)
Full FT
Full FT+L2-SP
LoRA-4
LoRA-16
OPLoRA-16
(a) Subspace interference ρ_k
ρ_k ($k = 128$)
0.15
0.10
0.05
0.00
k/d_{out}
0
0
0
layer 0
layer 13
layer 27
(b) Relative update size η
η (%)
1.5
1.0
0.5
0.0
layer 0
layer 13
layer 27
(c) Activation correction $R(z)$
$R(z)$ (mean over 32 probes)
GSM8K (math)
HellaSwag
0.125
0.100
0.075
0.050
0.025
0.000
layer 0
layer 13
layer 27